



CAPSTONE PROJECT

Smart Waste Management System

A digital solution for cleaner, smarter cities

Member 1
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Introduction & Problem

Member 2
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Objectives & Features

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Technology ,Architecture & Demo

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Nibraz

Advantages & Conclusion

The Problem We're Solving

Waste management is a major challenge in many growing cities.



Overflowing Bins

Public and residential bins fill up faster than they are emptied, spilling waste into streets.



Delayed Collection

Garbage trucks follow fixed schedules with no visibility into real, current demand.



Improper Disposal

Uncollected waste leads to pollution, foul odor, and public health issues for the community.



The Result

Pollution, health risks, and unclean surroundings for residents.

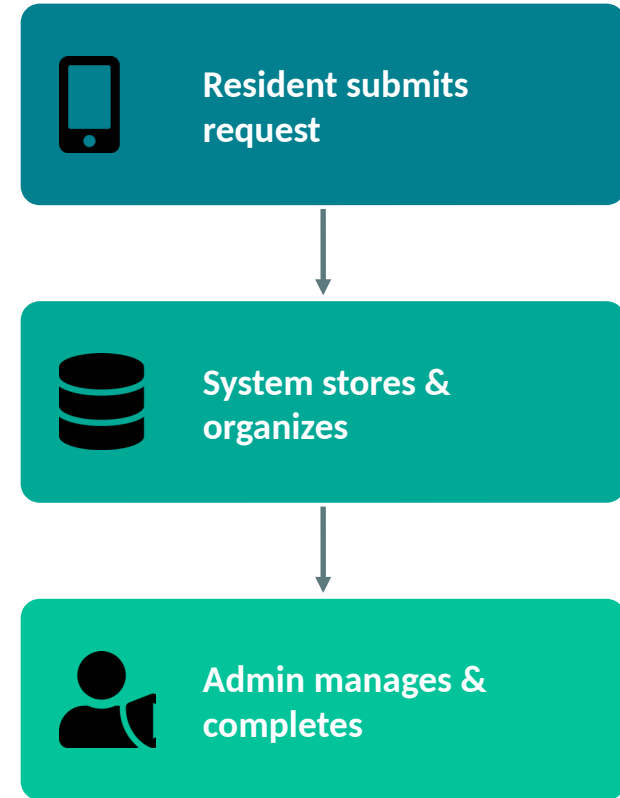
Our Fix

A simple digital system where users submit requests and admins manage & monitor collection efficiently.

Our Solution

A lightweight digital platform that connects residents and waste-management administrators. Residents submit collection requests in seconds; administrators track, manage, and resolve them from one organized system — keeping neighborhoods cleaner and services faster.

- ✓ Simple digital request submission
- ✓ Centralized admin monitoring dashboarding
- ✓ Faster, better-organized collection



Project Objectives

Simplifying the waste collection process through an easy-to-use system.



Simplify

Make requesting waste collection quick and effortless for every resident.



Automate

Replace manual, paper-based tracking with an organized digital workflow.



Improve Efficiency

Give administrators clear visibility to manage and resolve requests faster.

Key Features

Five core capabilities that power the system end-to-end.



Add Requests

Log a new waste collection request in moments.



View Pending

See every open request at a glance.



Mark Completed

Close out requests once waste is collected.



Search by Area

Filter and find requests by location.



Manage Records

Store and organize all data reliably.

Together, these features reduce manual work and make waste management faster and more organized.

Why It Matters

Turning five simple features into real operational impact.



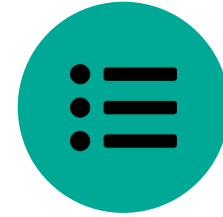
Less Manual Work

Digital requests replace paper logs and phone calls.



Faster Response

Admins see and act on pending pickups immediately.



Better Organization

Every record is searchable, tracked, and up to date.

Net effect: a cleaner, more responsive waste collection service for the whole community.

Technologies Used

Built in Java, using core object-oriented and collection concepts.



Classes & Objects

Model requests and system behavior.



ArrayList

Stores all collection requests in memory.



Scanner

Captures user input from the console.



Methods

Separate logic for each operation.



Loops

Iterate through stored requests.

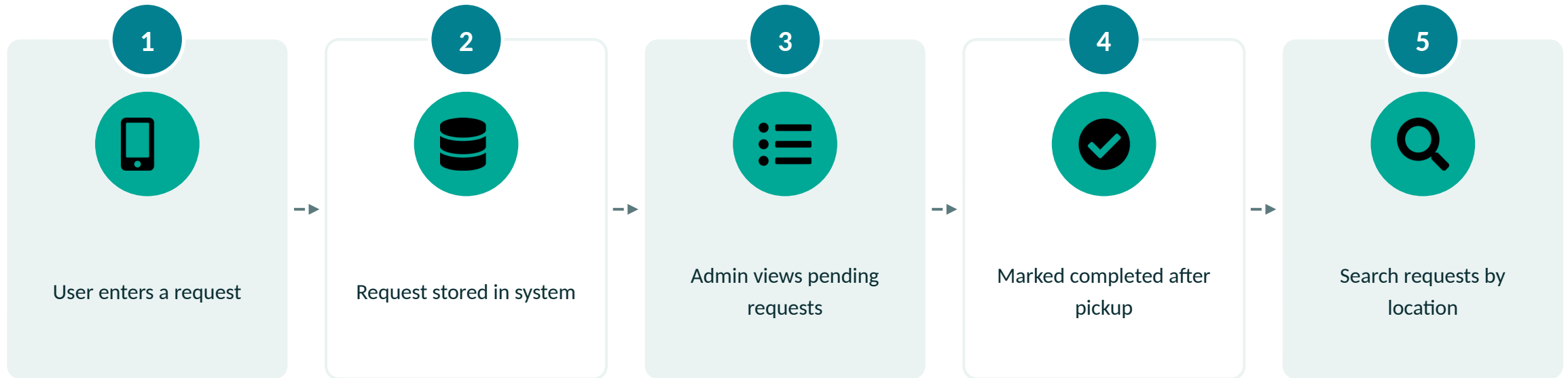


Conditionals

Drive decisions like status checks.

How It Works

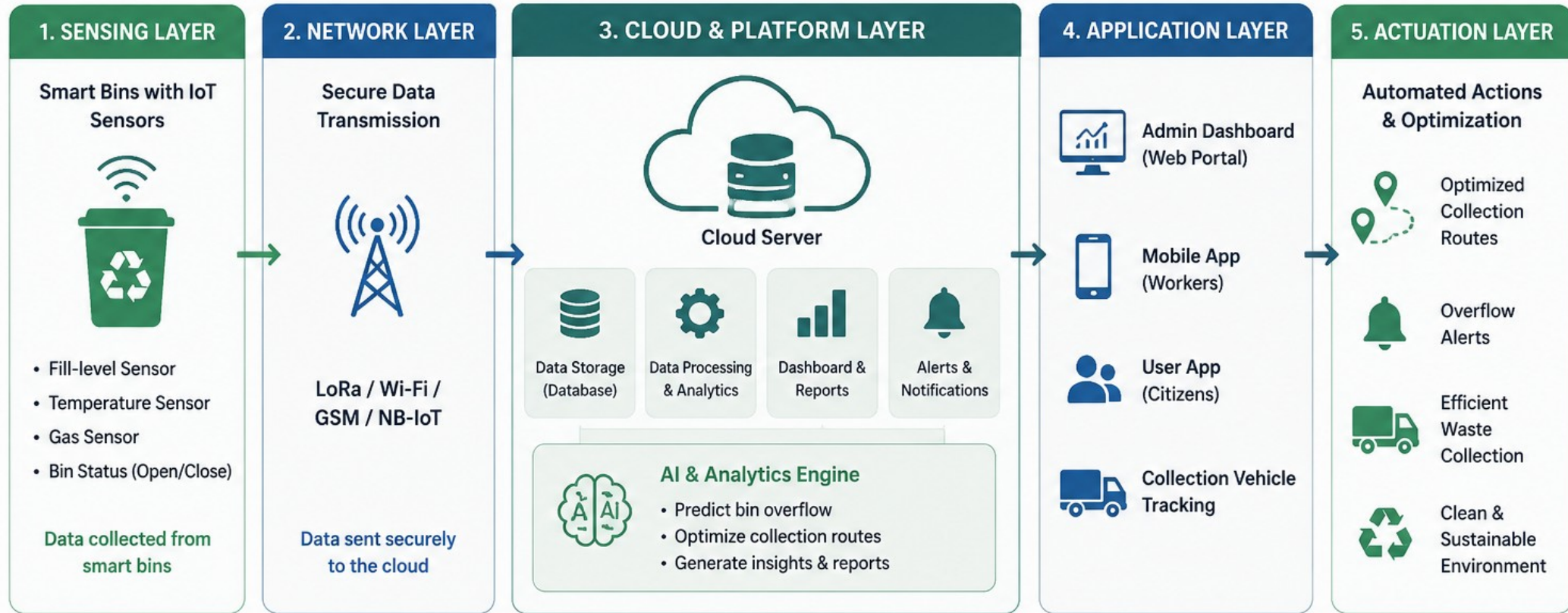
Five steps from request to resolution.



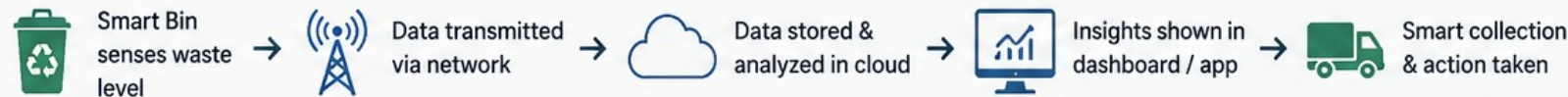
SMART WASTE MANAGEMENT SYSTEM – ARCHITECTURE

Blueprint of the System

An IoT and Cloud based architecture that enables real-time monitoring, efficient collection, route optimization and data driven decision making.



FLOW SUMMARY



Goal: Efficient Waste Management for a Cleaner Tomorrow

Live Demonstration

Walking through the running Java program.



```
> Add Waste Request
  Name: Priya | Area: Sector-4 | Type: Household
  ✓ Request added.

> View Pending Requests
  [1] Priya – Sector-4 – Household
  [2] Arjun – Lakeview – Recyclable

> Mark Completed
  Request [1] marked as COMPLETED.

> Search by Area: "Lakeview"
  [2] Arjun – Lakeview – Recyclable
```



Add a waste request

Enter name, area, and waste type.



Display pending requests

Review every open request in the list.



Mark one as completed

Update status once collected.



Search by area

Filter requests using the area name.

Result: simple, efficient, and easy to use.

Advantages & Future Scope



Advantages

- Easy to use
- Saves time
- Reduces paperwork
- Cleaner surroundings
- Improves collection management



Future Scope



Mobile application



GPS tracking for garbage trucks



SMS / email notifications



IoT smart bins that self-report

Alignment with the UN Sustainable Development Goals

How a Smart Waste Management System advances four SDGs at once.



SDG 11

Sustainable Cities & Communities

Digitally tracked collection reduces overflowing bins and delayed pickups, directly supporting cleaner, more livable, better-managed urban spaces.



SDG 12

Responsible Consumption & Production

Structured logging of waste type and status encourages proper segregation and disposal, supporting more sustainable management of resources.



SDG 13

Climate Action

Organized, on-demand collection cuts wasted truck trips today, and paves the way for GPS routing and IoT sensors that shrink emissions further.



SDG 3

Good Health & Well-Being

Faster resolution of collection requests limits uncollected waste, pollution, and the health hazards it creates for residents nearby.

By digitizing a basic civic service, this small-scale project models the same principles cities need to meet global sustainability targets.

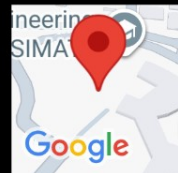


A Simple, Practical Step Toward Smarter Cities

Our Smart Waste Management System makes waste collection more efficient and supports cleaner, smarter communities.



Thank You — We're Happy to Answer Any Questions



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